These notes cover practical Data Engineering interview questions commonly asked in interviews for Freshers and candidates with 1–3 years of experience.

The focus is on understanding **how to approach real-world scenarios**, not just memorizing definitions.

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1. Design a Simple ETL Pipeline

Interview Question

A company receives a CSV file every morning and wants the data to be available in Power BI within an hour.

How would you design the pipeline?

Expected Pipeline Flow

Source CSV
↓
Landing Zone
↓
Data Validation
↓
Data Cleaning
↓
Business Transformations
↓
Data Warehouse / Delta Table
↓
Power BI Dashboard

Key Points

- Store raw data first.
- Validate the incoming data.
- Clean and transform the data.
- Load the final data into the reporting layer.
- Keep logging and monitoring throughout the pipeline.

Interview Tip

Always explain **why** each stage exists instead of simply listing the stages.

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2. Production Pipeline Failure

Interview Question

Your production pipeline failed at 2 AM.

What would you do?

Recommended Approach

1. Check the orchestration logs.
2. Identify the failed task.
3. Read the complete error message.
4. Verify source availability.
5. Check database and storage connectivity.
6. Fix the root cause.
7. Re-run only the failed step if supported.

Interview Tip

Do not answer:

> "I will simply restart the pipeline."

Interviewers expect a structured debugging approach.

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3. Handling Duplicate Records

Interview Question

How would you handle duplicate customer records?

Approach

- Understand the business definition of a duplicate.
- Identify the business key.
- Remove exact duplicates if appropriate.
- Keep only the latest record when timestamps are available.
- Maintain audit information if required.

Common Business Keys

- Customer ID
- Email
- Phone Number
- Composite Key

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4. Full Load vs Incremental Load

Interview Question

A table contains 100 million records.

Only 20,000 records change every day.

Which loading strategy would you use?

Full Load

Reloads the complete dataset.

Incremental Load

Loads only new or modified records.

When to Use Full Load

- Initial migration
- Complete data rebuild
- Major schema changes

When to Use Incremental Load

- Daily ETL pipelines
- Large production datasets

- Cost optimization

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5. Incorrect Business Report

Interview Question

Business reports incorrect revenue.

How would you investigate?

Recommended Approach

Trace the data backwards.

Power BI
↓
Gold Table
↓
Silver Table
↓
Bronze Table
↓
Source File
↓
Source System

Interview Tip

Avoid assumptions.

Validate every stage of the pipeline.

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6. Late Arriving Source File

Interview Question

The pipeline starts at 1 AM.

The source file arrives at 2 AM.

What should happen?

Possible Solutions

- Retry automatically.
- Wait for file arrival.
- Trigger event-based execution.
- Notify support team.
- Process the file once available.

Best Practice

Avoid hardcoded wait times.

Use configurable retry mechanisms.

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7. Medallion Architecture

Bronze Layer

- Raw data
- No transformations
- Historical backup

Silver Layer

- Cleaned data
- Standardized schema
- Duplicate removal
- Business validations

Gold Layer

- Reporting-ready tables
- Aggregations
- KPIs
- Business metrics

Why Use Medallion Architecture?

- Easier debugging
- Better data quality
- Simplifies reporting
- Supports data reprocessing

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8. Small File Problem

Interview Question

Why are millions of small files considered bad?

Problems

- High metadata overhead
- More Spark tasks
- Slower reads
- Poor query performance

Solutions

- File compaction
- Delta OPTIMIZE
- Better partitioning
- Controlled output file sizes

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9. Slow Spark Job

Interview Question

Yesterday the job finished in 15 minutes.

Today it takes 2 hours.

What would you investigate?

Checklist

- Data volume
- Data skew
- Shuffle operations
- Partition count
- Join strategy
- Executor memory
- Spark UI
- Cluster health

Interview Tip

Never recommend increasing cluster size as the first solution.

Always identify the root cause.

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10. Design an End-to-End Data Pipeline

Example Flow

Source Database
↓
Ingestion
↓
Bronze Layer
↓
Validation
↓
Transformation
↓
Silver Layer
↓
Business Aggregation
↓

Gold Layer
↓
Power BI

Additional Components

- Scheduling
- Logging
- Monitoring
- Retry Logic
- Error Handling
- Alerts

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11. Pipeline Logging

Information to Log

- Pipeline Name
- Start Time
- End Time
- Execution Status
- Number of Records Read
- Number of Records Written
- Failed Records
- Error Message
- Source File Name
- Execution Duration

Why Logging Is Important

Logging helps identify failures quickly and simplifies production debugging.

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12. Customer Count Doubled

Interview Question

Business reports that customer count has suddenly doubled.

How would you investigate?

Possible Reasons

- Duplicate source data
- Duplicate ingestion
- Incorrect joins
- Missing deduplication
- Pipeline executed twice
- Incorrect merge logic

Recommended Approach

Never assume the cause.

Investigate systematically across every stage.

Frequently Asked Theoretical Questions

ETL vs ELT

ETL

- Transform before loading.
- Common in traditional data warehouses.

ELT

- Load first.
- Transform inside modern cloud data platforms.

Why Use Parquet Instead of CSV?

- Columnar storage
- Better compression
- Faster reads
- Supports predicate pushdown

What Is Data Lineage?

Tracks the complete journey of data from the source system to the final report.

What Is Data Validation?

Verifying that incoming data meets business and technical expectations before processing.

Examples:

- Null checks
- Duplicate checks
- Data type validation
- Range validation

What Is Data Reconciliation?

Comparing source and target systems to ensure data consistency.

What Is Schema Evolution?

Allowing datasets to handle schema changes without rebuilding the entire pipeline.

Why Partition Data?

Partitioning helps Spark read only the required data instead of scanning the complete dataset.

Benefits:

- Faster queries
- Reduced I/O
- Better performance

What Is Idempotency?

An idempotent pipeline produces the same result even if it is executed multiple times.

This prevents duplicate data in production.

Batch Processing vs Streaming

Batch

- Processes data at scheduled intervals.
- Suitable for reports and historical processing.

Streaming

- Processes data continuously as events arrive.
- Suitable for real-time analytics and monitoring.